



Islamic University of Technology

Department of Computer Science and Engineering (CSE)

CSE 4205: Digital Logic Design

Summer, 2024–2025

Aashnan Rahman, Jr. Lecturer, CSE

Lecture 13

Designing Sequential Circuits

Contents

1	Designing Sequential Circuits	2
1.1	General Design Procedure	2
2	Two-Bit Synchronous Up Counter Using T Flip-Flops	3
2.1	Steps 1–3: Selection, Excitation Table, and State Diagram	3
2.2	Step 4: State/Excitation Table and Input Equations	3
2.3	Step 5: Logic Diagram	4
3	Two-Bit Synchronous Up Counter Using JK Flip-Flops	5
3.1	Steps 1–3: Selection, Excitation Table, and State Diagram	5
3.2	Step 4: State/Excitation Table and Input Equations	5
3.3	Step 5: Logic Diagram	6
4	Three-Bit Synchronous Up Counter Using T Flip-Flops	7
4.1	Steps 1–3	7
4.2	Step 4: State/Excitation Table and Input Equations	7
4.3	Step 5: Logic Diagram	8
5	Three-Bit Synchronous Up Counter Using JK Flip-Flops	9
5.1	Steps 1–3	9
5.2	Step 4: State/Excitation Table and Input Equations	9
5.3	Step 5: Logic Diagram	10
6	Practice Problems	11
7	Worked Example: A Custom State Sequence	12
7.1	Steps 1–3	12
7.2	Step 4: State/Excitation Table and Input Equations	12

1 Designing Sequential Circuits

Sequential circuits are fundamental building blocks in digital logic design. Their outputs depend on both the current inputs and the circuit's previous state. Unlike combinational circuits, sequential circuits contain memory elements such as flip-flops.

1.1 General Design Procedure

Step 1: Choose the number and type of flip-flops.

Step 2: Write the excitation table of the selected flip-flop.

Step 3: Construct the state diagram and combine it with the excitation table to form the state/excitation table.

Step 4: Derive and simplify the equations for the flip-flop inputs.

Step 5: Draw the logic diagram.

For a sequence containing states represented by values up to M , choose

$$n = \lceil \log_2(M + 1) \rceil$$

flip-flops, where n is the number of state bits.

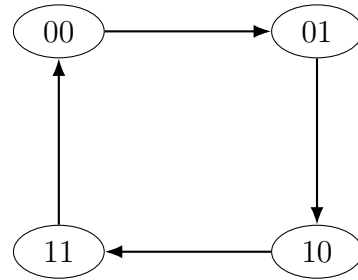
2 Two-Bit Synchronous Up Counter Using T Flip-Flops

2.1 Steps 1–3: Selection, Excitation Table, and State Diagram

Two T flip-flops are required. Their outputs are denoted by Q_A (most significant bit) and Q_B (least significant bit).

Q_n	Q_{n+1}	T
0	0	0
0	1	1
1	0	1
1	1	0

T flip-flop excitation table



State sequence: $00 \rightarrow 01 \rightarrow 10 \rightarrow 11 \rightarrow 00$

2.2 Step 4: State/Excitation Table and Input Equations

Present state		Next state		T inputs	
Q_A	Q_B	Q_A^+	Q_B^+	T_A	T_B
0	0	0	1	0	1
0	1	1	0	1	1
1	0	1	1	0	1
1	1	0	0	1	1

Table 1: State/excitation table for a two-bit synchronous up counter using T flip-flops.

T_A		
$Q_A \backslash Q_B$	0	1
0	0	1
1	0	1

T_B		
$Q_A \backslash Q_B$	0	1
0	1	1
1	1	1

Therefore,

$$\boxed{T_A = Q_B}, \quad \boxed{T_B = 1}.$$

2.3 Step 5: Logic Diagram

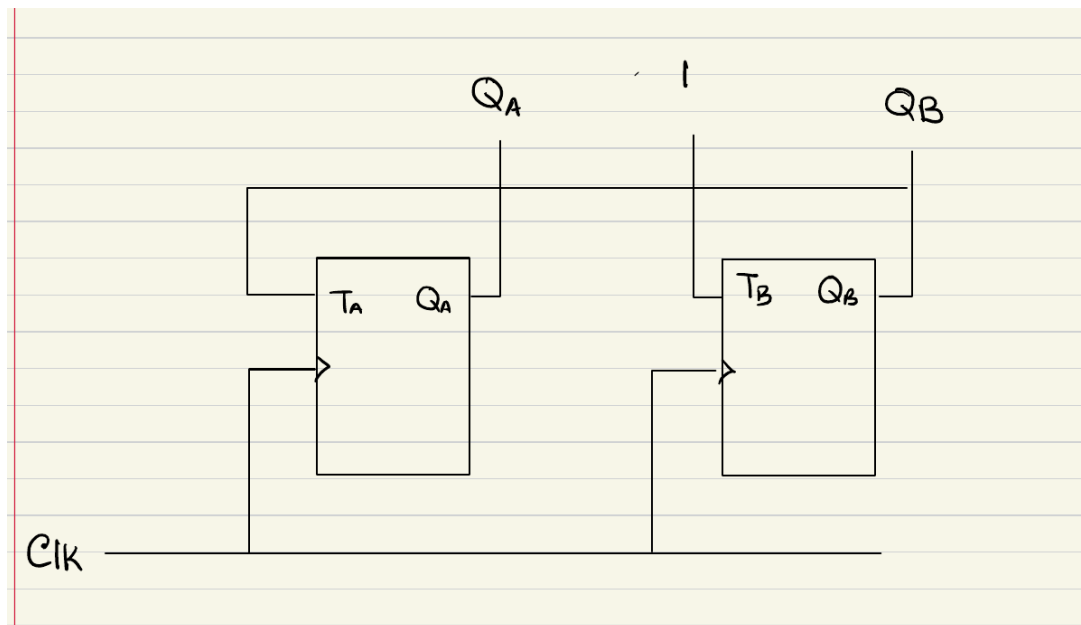


Figure 1: Two-bit synchronous up counter using T flip-flops (from the source notes).

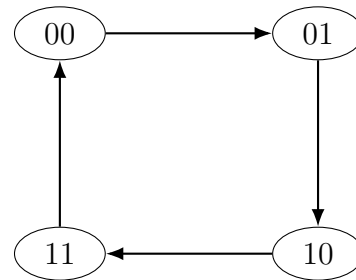
3 Two-Bit Synchronous Up Counter Using JK Flip-Flops

3.1 Steps 1–3: Selection, Excitation Table, and State Diagram

Two JK flip-flops are required. The state sequence is unchanged.

Q_n	Q_{n+1}	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

JK flip-flop excitation table



3.2 Step 4: State/Excitation Table and Input Equations

Present state		Next state		Flip-flop A		Flip-flop B	
Q_A	Q_B	Q_A^+	Q_B^+	J_A	K_A	J_B	K_B
0	0	0	1	0	X	1	X
0	1	1	0	1	X	X	1
1	0	1	1	X	0	1	X
1	1	0	0	X	1	X	1

Table 2: State/excitation table for a two-bit synchronous up counter using JK flip-flops.

J_A	
$Q_A \backslash Q_B$	
	0 1
0	0 1
1	X X
J_B	
$Q_A \backslash Q_B$	
	0 1
0	1 X
1	1 X

K_A	
$Q_A \backslash Q_B$	
	0 1
0	X X
1	0 1
K_B	
$Q_A \backslash Q_B$	
	0 1
0	X 1
1	X 1

Thus,

$$\boxed{J_A = K_A = Q_B}, \quad \boxed{J_B = K_B = 1}.$$

3.3 Step 5: Logic Diagram

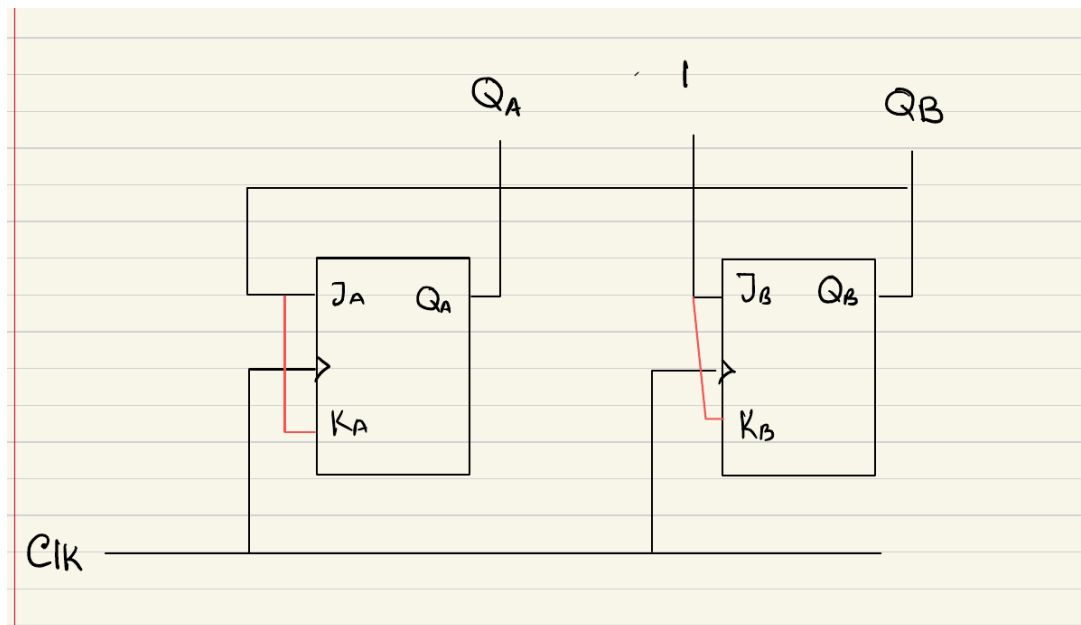


Figure 2: Two-bit synchronous up counter using JK flip-flops (from the source notes).

4 Three-Bit Synchronous Up Counter Using T Flip-Flops

4.1 Steps 1–3

Three T flip-flops are required. Let Q_A be the most significant bit and Q_C the least significant bit. The counter follows the eight-state cycle below.

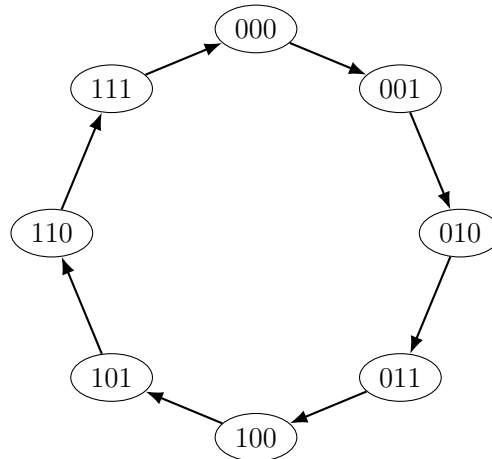


Figure 3: State diagram of a three-bit binary up counter.

4.2 Step 4: State/Excitation Table and Input Equations

Present state			Next state			T inputs		
Q_A	Q_B	Q_C	Q_A^+	Q_B^+	Q_C^+	T_A	T_B	T_C
0	0	0	0	0	1	0	0	1
0	0	1	0	1	0	0	1	1
0	1	0	0	1	1	0	0	1
0	1	1	1	0	0	1	1	1
1	0	0	1	0	1	0	0	1
1	0	1	1	1	0	0	1	1
1	1	0	1	1	1	0	0	1
1	1	1	0	0	0	1	1	1

Table 3: State/excitation table for a three-bit synchronous up counter using T flip-flops.

The Karnaugh maps are

		T_A						T_B						T_C			
$Q_A \backslash Q_B Q_C$		00	01	11	10	$Q_A \backslash Q_B Q_C$		00	01	11	10	$Q_A \backslash Q_B Q_C$		00	01	11	10
0		0	0	1	0	0		0	1	1	0	0		1	1	1	1
1		0	0	1	0	1		0	1	1	0	1		1	1	1	1

Hence,

$$T_A = Q_B Q_C, \quad T_B = Q_C, \quad T_C = 1.$$

4.3 Step 5: Logic Diagram

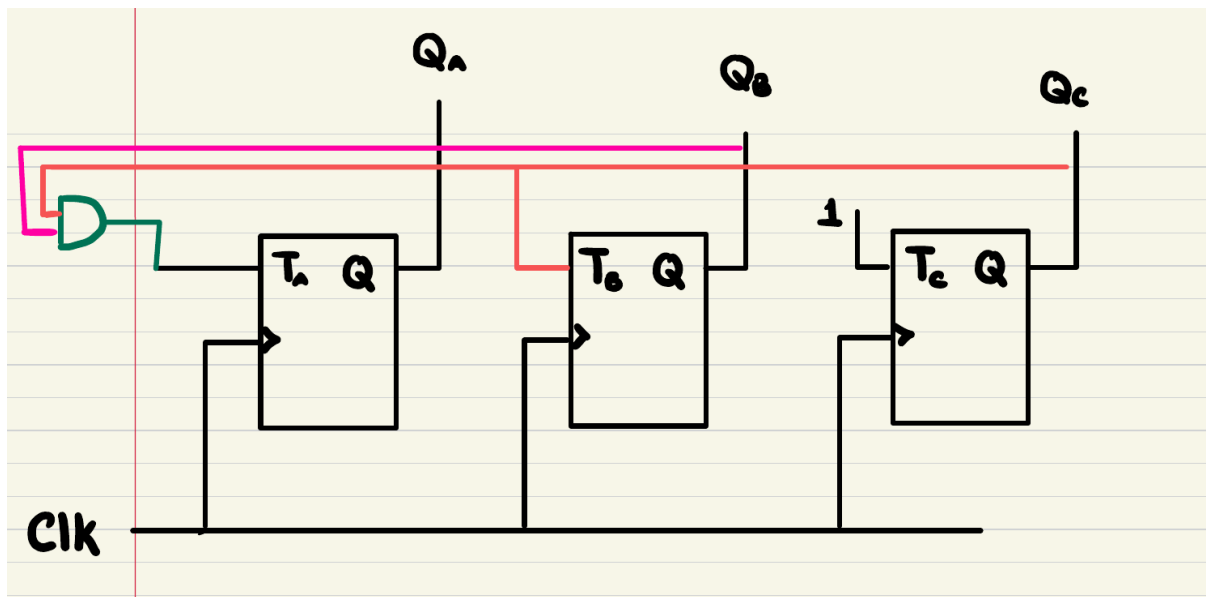
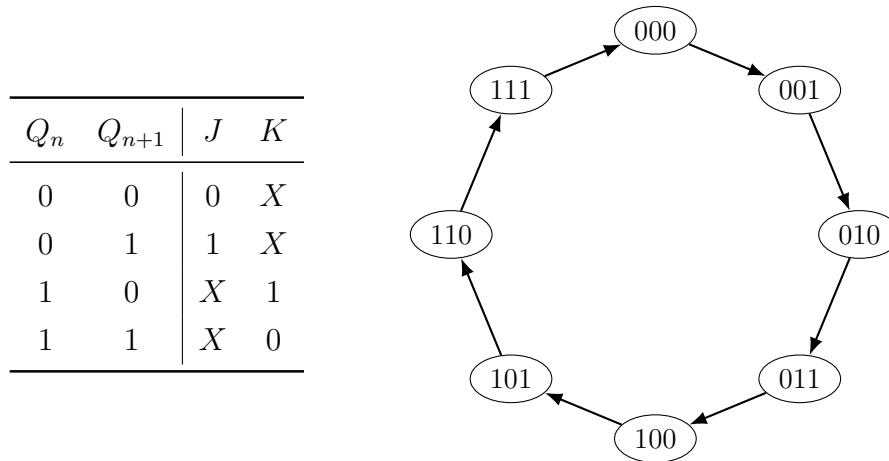


Figure 4: Three-bit synchronous up counter using T flip-flops (from the source notes).

5 Three-Bit Synchronous Up Counter Using JK Flip-Flops

5.1 Steps 1–3

Three JK flip-flops are required. The JK excitation table and the eight-state cycle shown previously are used.



5.2 Step 4: State/Excitation Table and Input Equations

Present state			Next state			A		B		C	
Q_A	Q_B	Q_C	Q_A^+	Q_B^+	Q_C^+	J_A	K_A	J_B	K_B	J_C	K_C
0	0	0	0	0	1	0	X	0	X	1	X
0	0	1	0	1	0	0	X	1	X	X	1
0	1	0	0	1	1	0	X	X	0	1	X
0	1	1	1	0	0	1	X	X	1	X	1
1	0	0	1	0	1	X	0	0	X	1	X
1	0	1	1	1	0	X	0	1	X	X	1
1	1	0	1	1	1	X	0	X	0	1	X
1	1	1	0	0	0	X	1	X	1	X	1

Table 4: State/excitation table for a three-bit synchronous up counter using JK flip-flops.

The Karnaugh maps for the six JK inputs are

		J_A			
$Q_A \backslash Q_B Q_C$		00	01	11	10
0		0	0	1	0
1		X	X	X	X

		K_A			
$Q_A \backslash Q_B Q_C$		00	01	11	10
0		X	X	X	X
1		0	0	1	0

		J_B			
$Q_A \backslash Q_B Q_C$		00	01	11	10
0		0	1	X	X
1		0	1	X	X

		K_B			
$Q_A \backslash Q_B Q_C$		00	01	11	10
0		X	X	1	0
1		X	X	1	0

		J_C			
$Q_A \backslash Q_B Q_C$		00	01	11	10
0		1	X	X	1
1		1	X	X	1

		K_C			
$Q_A \backslash Q_B Q_C$		00	01	11	10
0		X	1	1	X
1		X	1	1	X

Their simplifications give

$$\boxed{J_A = K_A = Q_B Q_C}, \quad \boxed{J_B = K_B = Q_C}, \quad \boxed{J_C = K_C = 1}.$$

The don't-care entries permit more than one valid Karnaugh-map cover. The expressions above are the minimal symmetric forms and lead directly to the standard synchronous-counter circuit.

5.3 Step 5: Logic Diagram

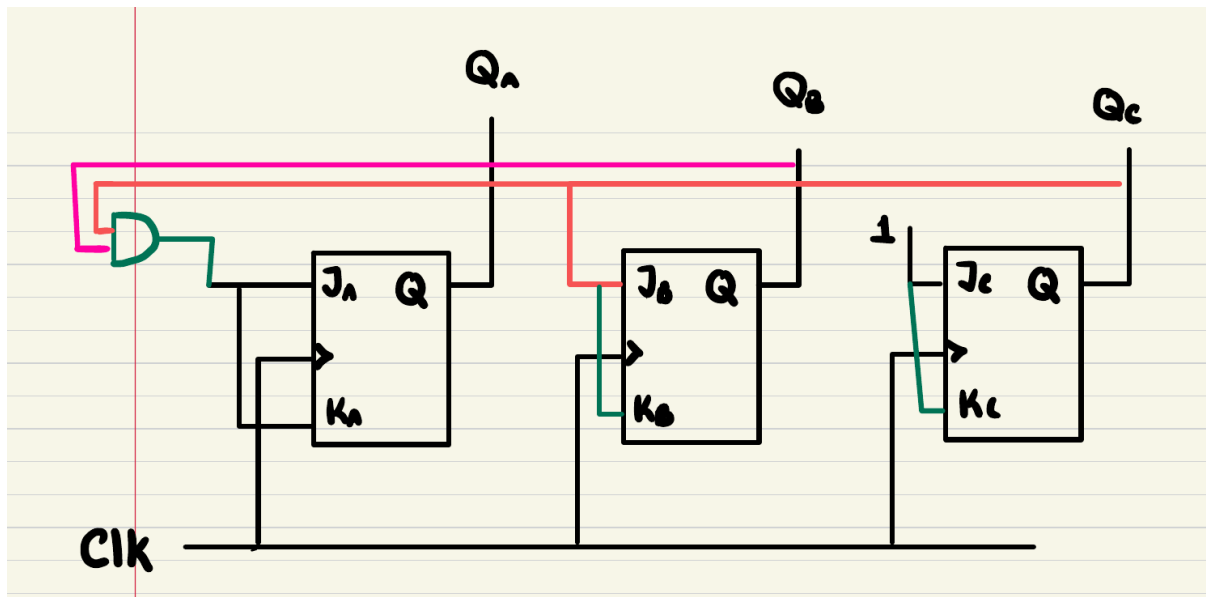


Figure 5: Three-bit synchronous up counter using JK flip-flops (from the source notes).

6 Practice Problems

Using the same design procedure, design the following circuits.

1. A three-bit synchronous down counter.
2. A synchronous modulo-12 up counter.
3. A synchronous sequential circuit that repeatedly follows the sequence

$$2 \rightarrow 7 \rightarrow 5 \rightarrow 1 \rightarrow 4 \rightarrow 2.$$

7 Worked Example: A Custom State Sequence

Design a synchronous sequential circuit using T flip-flops that follows

$$3 \rightarrow 2 \rightarrow 5 \rightarrow 4 \rightarrow 1 \rightarrow 7 \rightarrow 3.$$

7.1 Steps 1–3

The largest state value is 7, so three state bits are necessary:

$$n = \lceil \log_2(7 + 1) \rceil = 3.$$

Choose three T flip-flops with state outputs $Q_A Q_B Q_C$. The required state cycle is

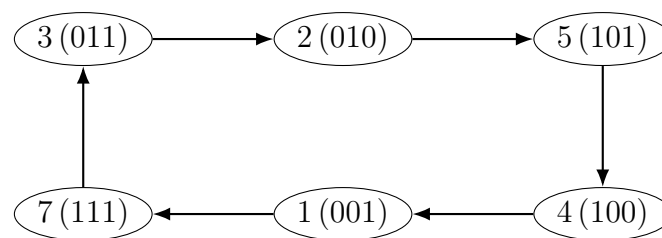


Figure 6: Required custom state sequence.

States 000 and 110 are unused and may be treated as don't-care states during minimization.

Important mapping note. When filling the Karnaugh maps, check the *input (present-state) side* of each transition in the state/excitation table. The states in this problem do not occur in numerical order, so the values must be placed in the K-map cells identified by the present-state bits $Q_A Q_B Q_C$, rather than copied in the order in which the transitions are listed.

7.2 Step 4: State/Excitation Table and Input Equations

Transition	Present state			Next state			T inputs		
	Q_A	Q_B	Q_C	Q_A^+	Q_B^+	Q_C^+	T_A	T_B	T_C
$3 \rightarrow 2$	0	1	1	0	1	0	0	0	1
$2 \rightarrow 5$	0	1	0	1	0	1	1	1	1
$5 \rightarrow 4$	1	0	1	1	0	0	0	0	1
$4 \rightarrow 1$	1	0	0	0	0	1	1	0	1
$1 \rightarrow 7$	0	0	1	1	1	1	1	1	0
$7 \rightarrow 3$	1	1	1	0	1	1	1	0	0

Table 5: State/excitation table for the custom sequence.

T_A					T_B					T_C				
$Q_A \backslash Q_B Q_C$	00	01	11	10	$Q_A \backslash Q_B Q_C$	00	01	11	10	$Q_A \backslash Q_B Q_C$	00	01	11	10
0	X	1	0	1	0	X	1	0	1	0	X	0	1	1
1	1	0	1	X	1	0	0	0	X	1	1	1	0	X

The simplified input equations are

$$T_A = \overline{Q_C} + \overline{Q_A} \overline{Q_B} + Q_A Q_B,$$

$$T_B = \overline{Q_A} \overline{Q_B} + Q_B \overline{Q_C},$$

$$T_C = Q_A \overline{Q_B} + \overline{Q_A} Q_B = Q_A \oplus Q_B.$$

These expressions can be connected to the T_A , T_B , and T_C inputs of three clocked T flip-flops to complete the logic diagram.

Try to construct the logic diagram yourself, and compare it with the solution.

For your convenience, the solution is provided in the [sequential circuit file](#).

If you simply toggle the clock, the circuit may not work initially because its default state is **000**, which is not a valid state for this circuit. First, set the circuit to a valid state, such as **101**.

To do this, briefly activate the **Set** pin of the required flip-flops by clicking it to 1 and then returning it to 0. Do **not** leave the Set pin at 1, as the flip-flop will remain set continuously.

Once the circuit is in a valid state, toggle the clock pulse and observe how the circuit changes state.